

Response to reviewer and editor comments

Manuscript: “A real-time reservoir inflow forecast evaluation framework” (MS 8299185), submitted to the *Journal of the American Water Resources Association*.

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We thank the editor and reviewers for their comments. The main concern was that the manuscript described a framework without identifying its contribution. We now define the contribution as the design of a multi-product operating trial. The procedure aligns competing forecasts over a common period, defines decisions and constraints before metrics are selected, uses regular operator review to revise the evaluation, and specifies how long the trial should continue. This setting includes cases where product-specific hindcasts are unavailable, but also cases where they cover different periods, model versions, or system configurations and therefore do not provide a comparable view of current performance.

We also expanded the descriptions of the system and forecast products, added a smoothing sensitivity analysis, characterized the evaluation period, and connected forecast quality to the forecast value literature.

The audit also identified four errors or qualifications that required correction:

- The evaluation period was given as July 2023 to May 2025. The record analyzed runs from June 2023 to April 2025.
- The three forecast products do not cover the period uniformly. Forecast B spans the full period; forecast A and the in-house forecast both end in November 2024. The metrics in Table 1 are therefore computed over overlapping periods but not identical sample sizes. We now report the sample size for each product and have verified that restricting the comparison to the 510 days on which all three are present does not change the ranking at any location.
- The paragraph comparing metrics across the three dams labeled the reported values as KGE when they were NSE. This is the passage Reviewer 1 asked us to check (comment 1.6), and rewriting it is the largest change to the results.
- The cumulative forebay calculation was incorrect. We have corrected it and aligned the forecasts so the cumulative error can be compared.

Reviewer comments are set in indented italics, trimmed to the operative text where they are long. A tracked-changes manuscript accompanies this response, so we quote revised text only where the wording is itself the answer.

Editor comments

The greatest barrier [...] is the question of novelty and reproducibility. Reviewers 1, 3, and 4 note that there is no distinct statement of novelty, with reviewer 3 particularly noting that the manuscript seems to suggest that the framework is the novelty without fully describing the novelty.

We revised the introduction to state the contribution directly. The procedure addresses how a utility should compare multiple products over the same current operating period when separate hindcasts are unavailable or not comparable. It now begins by defining the decisions, horizons,

constraints, incumbent forecast, and operating regimes that establish what performance means. All products are evaluated against the same observations. Operators then review results at regular meetings, identify errors that changed or could have changed a decision, and revise the metrics while preserving why each change was made. For a seasonal system, the trial continues for at least one annual cycle.

We also clarify how this trial complements a hindcast. The real-time evaluation identifies operationally viable products and records current constraints, data interruptions, operator adjustments, and decision rules. A subsequent common hindcast can then evaluate the selected products across a longer hydrologic record and estimate revenue, water supply, spill, and other outcomes using the same current system representation.

On reproducibility, we have anonymized the proprietary data and made the forecast evaluation as reproducible as we could given the limitations of the data. The evaluation code is public (<https://github.com/HydroWIREs-PNNL/inflow-forecast-evaluation>) and sample data are on Zenodo (<https://doi.org/10.5281/zenodo.16921728>). The repository now also includes a script that reproduces every number reported in the paper from the published data. We have added the parameter values that were previously left implicit, most importantly the smoothing window (comment 4.1), together with a sensitivity analysis. The proprietary forecast time series cannot be released, which we state as a limitation.

Reviewer 1

The topic and the contents are relevant, and it is very interesting for hydropower operators and decision-makers. However, the novelty and main contributions of this study are unclear.

We address the comments on framing, system description, tables, and figures below.

1.1 Novelty

The novelty and main contributions of this study are unclear. Please explain and discuss in detail.

Please see our response to the editor for the full statement of contribution. The contribution is not a new verification statistic or the use of operational data alone. It is a documented procedure for designing a comparable trial of multiple forecast products. The procedure defines the decisions and constraints before selecting metrics, holds the observation period and operating context common across products, specifies regular operator review and metric revision, records the reason for each change, and sets the evaluation duration from the operating regimes that must be sampled. This is needed even when product-specific hindcasts exist because separate hindcasts may describe different model versions, meteorological inputs, periods, infrastructure, or operating rules.

We have added to the discussion why this matters now. Forecast products using AI and machine learning methods are being marketed to utilities at an increasing rate, and the verification statistics vendors supply may not be verifiable. A utility generally has no standard procedure for checking those claims against its own system, and as we now report, often cannot inspect the model at all.

1.2 Persistence as a benchmark at long lead times

[...] Fig. 3 and Fig. 4 show that 10-day and months “persistence forecasts” as benchmark which is unrealistic in practice.

Persistence can be competitive at short lead times. We now recommend matching the benchmark to the decision horizon: persistence for hourly to two-day operations and climatology for weekly to

seasonal horizons. The captions state that persistence is retained beyond two days only to show its degradation and is not a meaningful long-horizon reference.

The figure captions now carry the same qualification.

Adding a climatological benchmark would be interesting but a defensible daily climatology for these locations requires a long record of calculated inflow at the dams. While there are nearby USGS gages, these reflect older operations and so would not give a representative climatology. For this reason we have not included it in the figures.

1.3 Operating problems and the forecasting scheme in practice

What are the main operation problems of these hydropower stations? Which forecasting scheme is used in practice?

We have added a subsection, “Operating context.” The three dams have limited storage relative to their inflow and are operated as run-of-river facilities, with a daily cycle of an upward ramp, flexible generation, a downward ramp, and refill. Generation is bid into the ISO New England day-ahead market, so the day-ahead inflow volume is the quantity that matters most. The binding constraint comes from the FERC relicensing of the three lower Connecticut River dams, which imposed a ± 0.5 foot (≈ 15 cm) limit on forebay fluctuation with penalties for exceedance. The central operating problem is committing generation a day ahead on an uncertain inflow forecast without violating the forebay limits, which is what motivated the cumulative-error and threshold-exceedance metrics.

On the scheme used in practice: the operators used an in-house forecast built from National Weather Service River Forecast Center forecasts, routed to the dam locations and adjusted using operator judgment. This is the incumbent against which the two commercial forecasts were evaluated, labeled GRH in the results, and it extends only about 32 hours because the day-ahead bid is the only decision it is used for.

1.4 Description of the forecast products

Two commercial forecasts (A, B) and the GRH in-house forecast (GRH) should be described in detail, such as what type of hydrologic model used? How about the calibration and validation results, what type input data requirement etc.

Reviewers 2 and 4 raised the same point (comments 2.16 and 4.3). We agree that anonymization without any methodological characterization leaves readers unable to interpret the results. We have added a subsection describing what we are able to disclose:

“Forecast A is produced by a consulting firm using a conceptual rainfall-runoff model with hydraulic routing through the Connecticut River main stem, driven by a commercial numerical weather prediction forecast, and issued once daily at an hourly time step out to roughly 8 days. Forecast B is produced by a technology firm using a machine learning model trained on gauge and remotely sensed catchment observations, issued twice daily at an hourly time step out to 10 days. The GRH in-house forecast combines RFC forecasts with routing and operator judgment and extends only through the day-ahead period, roughly 32 hours, because that is the horizon it is used for. Neither vendor released model internals or calibration and validation diagnostics, and we did not have access to them. A utility evaluating commercial forecast products generally cannot inspect them and must judge them on operational performance, which is part of the motivation for this framework.”

The issue frequencies and horizons are taken from the forecast archive rather than vendor documentation, so they describe what was actually delivered.

We cannot provide calibration and validation results for these models. The data used in this study was provided on the basis that it be anonymized. The lack of detailed information about the models reflects the reality that utilities face when procuring commercial forecasts. We have also added that a full procurement evaluation should weigh ongoing operational costs, data dependencies, and ease of integration alongside skill, since these may differ substantially between commercial products.

1.5 Watershed information

[...] *The watershed information, such as area, river length etc. should be described in detail.*

Agreed. We have added a table of physical characteristics for all six evaluation points, with drainage areas from the USGS for the reference gage at each location:

Location	Type	USGS reference gage	Area (mi ²)	Area (km ²)
Wilder	Mainstem dam	01144500 West Lebanon, NH	4,092	10,598
Bellows Falls	Mainstem dam	01154500 North Walpole, NH	5,493	14,227
Vernon	Mainstem dam	01156500 Vernon, VT	6,266	16,229
White River	Tributary	01144000 West Hartford, VT	690	1,787
Sugar River	Tributary	01152500 West Claremont, NH	269	697
Ottawaquechee River	Tributary	01151500 North Hartland, VT	221	572

We have also added text describing the cascading system. The mainstem areas are nested, so the area above Vernon includes those above Bellows Falls and Wilder. Because releases from each dam become inflow to the next, an error in an upstream forecast propagates to the downstream forecast and compounds with lead time, which we now quantify: for forecast A, day-ahead PBIAS at Bellows Falls is 0.4% and grows to -19.1% by day 7.

Basin hydrology is snowmelt influenced, with a spring freshet typically spanning March through May, convective rainfall in summer, and rain and rain-on-snow driven events in winter. Several tributaries have upstream flood control regulation by the U.S. Army Corps of Engineers.

1.6 NSE values in Table 1

In Table 1, the NSE values of forecast scheme A at Wilder, Bellows and Vernon watersheds are 0.72, 0.55, 0.81, respectively. Please check the evaluation results. The forecasting accuracy at the Bellows might high than that at the Wilder.

We recomputed the values; 0.72, 0.55, and 0.81 are correct. The accompanying text was not: it called these values KGE and has been corrected.

Forecast A does not have smaller errors at Bellows Falls than at Wilder. Its RMSE is 5,904 cfs at Bellows Falls and 2,278 cfs at Wilder. The apparent discrepancy arises because NSE is normalized by observed variance and should not be used by itself to rank accuracy across sites. The persistence rows make the site dependence visible: persistence NSE is -0.81 at Wilder, 0.39 at Bellows Falls, and 0.80 at Vernon.

We now compare products primarily within a location and interpret NSE together with RMSE, sample coverage, and the benchmark rows. We also repeated the calculation over the 510 days common to all products; the within-location product rankings do not change. An NSE skill score considered during revision was removed because the product and reference samples were not identical. A reference skill score is valid only when both forecasts use the same observations.

1.7 Table format

The format of Tables in the manuscript are poorly designed and needed be changed.

Table 1 has been rebuilt with rows grouped by location, the location named once per group, numeric columns aligned, consistent significant figures, and units in the column headers. It has gained three columns: the number of day-ahead forecasts evaluated for each product, which is necessary because the products cover different portions of the period; and RMSE in cfs (comment 4.7). Three tables have been added, giving basin characteristics (1.5), the smoothing sensitivity analysis (4.1), and the provenance of each tailored metric (3.2).

1.8 Figures

The figures also need be redrawn.

We revised all captions to state the quantity, units, and reference lines. We also redrew Figures 5 and 6 after correcting the cumulative-error calculation. The revised figures use three location panels, a common product palette, larger labels, ribbons for the central 95% interval in Figure 5, and percentages in Figure 6. Both now plot hours into each product’s delivered horizon rather than absolute lead time, and end at 32 hours; the captions state the absolute lead times each product’s window spans. Figure 6 now contains only the supported neutral-start stress test.

Reviewer 2

The core work is solid and the practical contribution is genuine. Assuming the major comments are addressed I recommend this manuscript is accepted with minor revision.

Thank you for a detailed and technically careful review. Several of the minor comments identify errors in the methods, including one in an equation. Both major comments have changed the manuscript substantially.

2.1 Engagement with the forecast value literature

The stated goal of the framework is to help operators evaluate whether a forecast product will improve operations enough to justify investment. The paper does not include any assessment of forecast value [...] The authors do not need to perform a full value assessment, but they should reference this literature, discuss how it connects to their framework, and provide operators with at least a conceptual pathway from the metrics presented to the investment decision the paper is motivated by.

We now distinguish quality from value in the abstract, introduction, and discussion. Quality describes forecasts relative to observations. Value also depends on the decision, its consequences, and risk preferences. We cite Thompson and Brier (1955) for cost-loss utility, Laugesen et al. (2023) for relative utility value, and Farkas et al. (2025) for a broader review.

We also give a GRH-specific pathway. The quality evaluation first screens products and creates an archive of forecasts as issued. Candidate forecasts can then drive a scheduling and dispatch model containing reservoir constraints, unit performance curves, day-ahead and real-time prices, and the incumbent operating strategy. Benefits should be measured relative to the forecast and decisions currently used. The forebay metric is decision-relevant but remains a stress test because it omits schedules and operator response.

2.2 Positioning of the contribution

The five-step iterative process [...] reads more as standard good operational practice than a novel framework contribution. I feel that the tailored metrics and operator feedback loop are where the genuine contribution lies, especially in the real-time context, and these deserve

more prominence and deeper treatment.

We restructured the manuscript around this distinction. Please see our response to the editor for the revised statement of contribution.

The revised process now has seven steps. The new first step requires operators to define the decisions, horizons, constraints, incumbent forecast, and operating regimes before metrics are selected. Products are then evaluated on common observations over an overlapping period. Soliciting feedback and acting on it remain separate as “Gather feedback” and “Update metrics,” because collecting operator comment and changing the metric set are different activities. The mechanics are documented through the meeting frequency, the questions asked of operators, the minimum duration for a seasonal system, and a table tracing each tailored metric to the operator concern that prompted it (3.2). We also explain how the real-time trial and a common hindcast form two stages of one evaluation rather than competing alternatives (3.4).

2.3 Hyphenation of “real time”

(L2, Title) “real time” should (perhaps) be hyphenated as “real-time” when used as an adjective (also applies throughout the manuscript).

Corrected throughout, including the title. We use “real-time” attributively and “in real time” adverbially. Reviewer 4 raised the same point in comment 4.5.

2.4 Verification improving alongside forecasts

(L12–13) The sentences seem to imply that we should expect verification methods to improve alongside improvements in forecasts, but this isn’t the case. [...] Suggest restructuring so the research gap is clearer.

You are right on both counts and the implication was unintentional. The passage has been rewritten so the gap is the subject: forecast quality and scheduling have improved while verification practice has not; traditional verification requires hindcasts over a climatologically representative period, often 30 years or more, and their cost and data requirements are why full verifications are not done in practice; the fallback of evaluating archived operational forecasts fits poorly with products that are updated continuously, trained on evolving data streams, hand adjusted by operators, or developed by third parties. We now also explain the research gap itself, which is the absence of a procedure by which operators can evaluate forecasts on their own system under familiar operating conditions.

The point about products delivered as services rather than inspectable models is now supported later in the paper, where we report that neither vendor released model internals or calibration diagnostics.

2.5 Dismissal of ensemble metrics

(L42–44) [...] the dismissal of ensemble evaluation metrics seems premature given that the authors later suggest probabilistic forecasts may help (L170). Consider briefly noting what ensemble metrics could be added if ensemble forecasts were available.

Agreed, and the inconsistency is a fair point. We set probabilistic evaluation aside early and then invoked probabilistic forecasts as a remedy later. Reviewer 4 asked for the same addition in comment 4.4, so we have added a methods subsection covering CRPS, rank histograms, and reliability diagrams; please see 4.4 for the text.

We have also softened the framing so that the focus on deterministic forecasts is presented as a

property of the products under evaluation rather than a judgment about ensemble metrics.

2.6 The argument against hindcast evaluation

(L45–47) The argument against hindcast evaluation is reasonable but somewhat overstated. Hindcast evaluations do not necessarily require operators to re-operate the system, especially if verifying forecasts of environmental variables such as inflows, which you are.

Agreed. Verifying a hindcast of inflow against observed inflow does not require re-operating the system. We have removed any suggestion that the real-time framework replaces a hindcast. The revised manuscript instead explains the comparability problem. Separate product hindcasts may use different periods, model versions, meteorological inputs, infrastructure, or operating rules. A common hindcast is valuable when these elements can be held constant, especially when the forecasts are coupled to an operations model to estimate revenue, water supply, spill, and other outcomes.

The real-time trial contributes a common snapshot of current performance and records delivery failures, operator adjustments, current constraints, and the decisions affected by forecast errors. It identifies the products and system representation that should enter a subsequent common hindcast. The approaches are therefore complementary rather than alternatives.

Additionally, the calculated inflows we verify against are derived from forebay elevation changes and dam outflow, so the observation record is itself a function of how the system was operated. This does not require re-operation to interpret, but it does mean our reference data is not independent of operations in the way a gauge record would be. We have added this to the limitations.

2.7 Persistence as “arguably worst possible forecast”

(L71) Describing persistence as “arguably worst possible forecast” is incorrect. [...] Suggest rephrasing to “a simple but competitive benchmark.”

Corrected using your suggested wording; see comment 1.2 for the revised passage.

Persistence has a day-ahead NSE of 0.80 at Vernon, confirming that it is a competitive short-horizon reference at that site. We no longer describe it as a lower bound.

2.8 Smoothing window width

(L80–81) A triangular smoother is reasonable but the choice of window width is not discussed. [...] Please note the window used and perhaps the sensitivity to this choice on your findings.

Reviewer 4 raised the same point in comment 4.1. We apply two successive 24-hour moving averages. Their convolution produces a triangular kernel spanning 47 hourly observations; “24-hour” denotes each component window. The component width was chosen to match the day-ahead operational timescale. Metrics improve modestly with window width, the largest change being 0.05 NSE, and the ranking of the three products is unchanged at every width at every location. Please see comment 4.1 for the table.

2.9 KGE name and citation

(L92) “King-Gupta Efficiency” should be “Kling-Gupta Efficiency.” Also, “(refs)” is a placeholder that needs to be replaced with actual references.

Both corrected, with Gupta et al. (2009) as you surmised, and Nash and Sutcliffe (1970) added for NSE in the same sentence. Reviewer 4 flagged the same two items in comment 4.6.

2.10 The claim about KGE and large values

(L92–93) The claim that KGE is not skewed by large values while NSE and RMSE are requires clarification. KGE and NSE both use correlation as a component. The relevant difference is that KGE decomposes performance into correlation, variability ratio, and bias ratio rather than using squared error. The statement as written is misleading.

You are right, and your characterization is the correct one. The original sentence attributed the difference to the squared error term, but KGE also incorporates correlation, which is itself computed from squared deviations, so the stated reason does not hold. We have replaced it with:

“We recommend KGE because it decomposes forecast performance into correlation, a variability ratio, and a bias ratio, so that a low score can be attributed to a specific deficiency rather than only registered as large aggregate error. NSE and RMSE aggregate squared error into a single number, which makes them more strongly influenced by the largest errors and less diagnostic about their source.”

We also state that KGE, NSE, RMSE, bias, and correlation answer different questions and should be interpreted together with sample coverage and benchmark forecasts.

2.11 Placement of the Knoben et al. citation

(L95–96) The KGE threshold of -0.41 is attributed to comparing against the mean of observations but the Knoben et al. (2019) citation is on L97, not adjacent to the claim.

Corrected. The citation now sits with the threshold statement it supports, followed by the consequence, previously left implicit: because the threshold is not zero, NSE and KGE should not be used interchangeably and intuition built on NSE values does not transfer to KGE. Knoben et al. (2019) also bears directly on the cross-location comparability point raised in comment 1.6, so we cite it in the benchmark section as well.

2.12 Bias versus error

(L98–99) “The bias of a single forecast point is the difference between observed and forecasted values” defines error (sometimes called residual), not bias.

Corrected. The text now reads:

“The error, or residual, of a single forecast is the difference between the forecast and the observation. Bias is the systematic component of error, the tendency of a forecast to be too high or too low on average, and is conventionally expressed for a set of forecasts as a percentage of total observed volume.”

The original definition also had the sign backwards relative to the PBIAS equation that follows it. The revised wording is consistent with the equation.

2.13 Pearson correlation formula

(Eq. 3) The Pearson correlation coefficient formula is missing a summation over the second term of denominator.

Corrected, and thank you for checking the equations. The denominator was written as a single summation over the product of squared deviations, and now reads as the product of two separate sums:

$$r = \frac{\sum(o_i - \mu_o)(f_i - \mu_f)}{\sqrt{\sum(o_i - \mu_o)^2 \sum(f_i - \mu_f)^2}}$$

We have checked the remaining equations; the NSE equation is added; the KGE and PBIAS expressions are correct as published. The reported correlation values were computed with a standard library function and are unaffected by the typesetting error.

2.14 Nominal capacity

(L107) 589 megawatts of “nominal” hydropower capacity. Is this installed capacity or average generation?

It is installed nameplate capacity. “Nominal” was ambiguous and has been replaced with “589 megawatts of installed nameplate hydropower capacity across 13 generating stations,” which also makes clear that the system total covers many more facilities than the three evaluated here.

2.15 Relationship among the three sites

(L109) Add a sentence describing how these 3 sites are related [...] Also good to note why you chose these sites over all the other sites (green dots in the figure).

Agreed. Your inference is correct: the three dams lie in series on the Connecticut River mainstem, Wilder upstream, then Bellows Falls, then Vernon, and are operated jointly, with releases from each becoming inflow to the next. This is why the drainage areas are nested and why bias propagates downstream, an effect we noted in the results without giving the reader the information needed to understand it. We now quantify the propagation (1.5).

On site selection, these three are the largest generators in the system, they are the dams subject to the FERC forebay constraint that motivated the tailored metrics, and they are where the operators concentrate their scheduling attention. The three tributaries were included because they are the only significant local inflows, entering between Wilder and Bellows Falls. The remaining facilities are smaller and not covered by the same license conditions. Both points are now in the text.

2.16 Anonymization and reproducibility

(L117) Anonymising the commercial forecast products is understandable but limits reproducibility. Consider at minimum describing the general class of model used by each [...] so readers can contextualise the results.

Agreed. Reviewers 1 and 4 asked for the same thing; please see comment 1.4 for the text we have added and for what we cannot disclose. We have also added the inability to inspect commercial products to the limitations, since it constrains any evaluation of this kind and is part of what motivates judging products on operational performance.

2.17 SI units

(Figure 1) Y-axis units are “cfs” (cubic feet per second). Consider also providing SI units for international readers.

Added. The Figure 1 caption gives the conversion ($1,000 \text{ cfs} \approx 28.3 \text{ m}^3/\text{s}$), and SI equivalents appear at first use in the text, including the 0.5 foot forebay limit as approximately 15 cm and the largest observed inflow in both units. We have kept cfs as the primary unit because it is what the operators and the FERC license use, and the tailored metrics are in feet of forebay for the same reason. We have not added a secondary axis to the figure itself; if the editor prefers that we will

do so.

2.18 Seasonal performance

(L134) “Forecasts tend to perform worse in the winter which is the wet season and worse in the summer which is the dry season.” This says forecasts perform worse in both wet and dry seasons. Please clarify when forecasts actually perform well and why.

You are correct, this was mixing up streamflow vs precipitation. Mean inflow at Vernon is highest in March and April and lowest from September through November, so the high-flow season is spring, driven by snowmelt, and that is where skill is highest for all three products.

The rewritten passage states this, and explains the mechanism: performance is poorest in the seasons where inflow is driven by precipitation events, rain and rain-on-snow in winter and convective storms in summer, both of which depend on highly uncertain precipitation forecasts, whereas snowmelt-driven inflow is comparatively predictable because it is governed by an accumulated snowpack observable in advance that melts over weeks, so skill depends more on temperature, which is forecast more reliably.

Two further corrections came out of this. The original text gave the snowmelt season as April through July, which does not match the monthly inflow pattern; it is now March through May. And the January and February KGE values for forecast A were given as 0.39 and -0.40 ; recomputing gives 0.45 and -0.67 . The revised value is also consistent with the figure, in which the February point falls below the plotted range, which -0.40 would not.

2.19 The inflow-equals-outflow assumption

(L146) [...] the forebay exceedance analysis assumes an inflow=outflow condition which is operationally unrealistic as the authors note. It would strengthen the analysis to discuss how operators actually deviate from this assumption and what the real-world exceedance rates might be.

Operators do not hold pass-inflow conditions. They schedule generation to a market bid, depart from inflow during the day, recover the forebay in off-peak hours, and intervene to keep the forebay in the operating band. We now call the calculation an idealized stress test and readily acknowledge that the system would never be operated in this way. It is neither a prediction nor a bound on realized compliance rates because operator actions can increase or decrease the modeled risk. Estimating realized rates requires schedules, constraints, and a model of operator response, as described under comment 4.9.

2.20 Extremes and the record length

(L178–182) [...] Consider also noting the limitation that almost two years of data may not capture the full range of extreme events which are often the conditions where forecast quality matters most for operational decisions.

Agreed, and we have stated the limitation explicitly. Reviewer 4 asked in comment 4.2 what the period did contain, so we addressed both together; please see that response for the events observed and their position in the long-term annual peak record. In short, the record covers a useful range but not the upper tail, and no severe multi-year drought.

One point specific to this system is worth adding: representing multiple high flow events matters less at GRH than it might elsewhere, because operational flexibility is limited during high flows and operators discount forecast errors in those conditions. The gap that does matter for GRH is

the absence of a strong multi-year drought, which we state as a limitation.

Reviewer 3

However, proposed framework itself is not well described, such as what is new, different from existing ones, and comparison of past studies are insufficient.

Our response to the editor describes the revised statement of contribution. Comments 3.4 and 3.5 are addressed below.

3.1 “market prices”

(l47) “market prices” — Could you please clarify what it means?

We mean day-ahead and real-time wholesale electricity prices, which affect generation bids and release schedules. Historical prices can often be retrieved. The harder quantities to reconstruct are outages, operating constraints, operator judgment, and how price signals changed decisions. We corrected this overstatement and now apply the reconstruction problem to a value assessment, not to direct verification of inflow quality.

3.2 How feedback is reflected in the process

(§2.1) Step 1 through 3 are so general. I assume that steps 4 and 5 are unique points in this manuscript. Please provide a more detailed explanation of how feedback is reflected in step 4.

You have identified the structure of the contribution correctly. We have now made the full trial design explicit rather than locating the contribution only in the later feedback steps.

The process now begins with “Define decisions and trial scope,” followed by collection of aligned real-time data, initial metrics, and visualization. “Gather feedback” and “Update metrics” remain separate, and the seventh step repeats the review while accumulating a sample that covers the operating regimes of interest.

Both steps are expanded with the mechanics. Feedback was collected in recurring meetings, monthly for most of the evaluation, each reviewing the current metrics against recent operating experience, with operators asked which recent forecast errors mattered operationally and which did not. Their answers frequently diverged from what the generic metrics indicated, especially in the high-flow season: they discounted large errors during spill conditions, when they had little discretion over releases and forecast quality was operationally irrelevant, while treating moderate errors with the potential to breach the forebay compliance band as serious. That is what motivated the tailored metrics, the exceedance metrics in particular.

We have also added a table to the manuscript tracing each tailored metric to the operator concern that prompted it and the approximate month it was added, so the process is auditable rather than asserted: day-ahead volumetric skill, because market bids are set from next-day volume rather than hourly shape (months 1–2); the seasonal breakdown, because products were suspected to behave differently in snowmelt and rain driven regimes (3–4); cumulative forebay error, because average error does not indicate whether the band would be breached (4–6); and forebay exceedance probability, because operators needed to know how quickly cumulative error could exhaust the operating band (5–6).

We should be candid that the months are approximate: the evaluation predates this paper and we kept no formal log of metric provenance, so these are reconstructed from meeting materials and

correspondence, though the sequence and motivations are accurate. A practical recommendation follows, which we now state: an evaluation of this kind should log metric provenance as it goes, since the record of why each metric exists is part of what makes the results defensible later.

3.3 Is a one-year learning period required?

(“5. Iterate”) This implies that at least one year learning period is required. Is this understanding correct?

The appropriate duration depends on the system. In this case metric development was complete within roughly six months, after which the evaluation accumulated data. Because the products show strong seasonal variation, we recommend at least one annual cycle for this application. That recommendation is not a universal minimum.

Our own results bear on the second point. Pooled day-ahead KGE for forecast A ranges from -0.67 in February to 0.88 in May, so an evaluation conducted in spring alone would have reached a substantially more favorable conclusion about that product than the full record supports. We have also noted the tension the operators raised, that industry preference runs toward shorter evaluations, sometimes a single season, while a single season samples only one hydrologic outcome.

3.4 Comparison with traditional methods

(result of case study) Were all the results derived through the proposed framework? If so, please compare those result with those obtained via “traditional” methods.

All results were produced through the framework, and we agree the comparison you describe is what demonstrates its effectiveness. We have added it in two forms.

First, the revised “Comparison with retrospective forecast evaluation” subsection now states what each approach contributes. A common hindcast provides larger samples, more extremes, and a basis for estimating revenue, water supply, spill, and other outcomes through an operations model. A real-time trial evaluates all products against the same current observations and records delivery failures, operator adjustments, current constraints, and the decisions that forecast errors affect. Separate vendor hindcasts may not be comparable if they use different periods, model versions, meteorological inputs, infrastructure, or operating rules.

The takeaway is a sequence. The real-time trial identifies operationally viable products and the constraints and decision rules that a value study must represent. The selected products can then be evaluated through a common hindcast and scheduling model over a longer record. If comparable hindcasts already exist, the two evaluations can proceed in parallel.

Second, the comparison of conclusions. A conventional verification of these forecasts, computing NSE, KGE, PBIAS, and correlation over the full record, concludes that forecast B outperforms forecast A and is comparable to the in-house forecast. That conclusion is correct and this framework reproduces it. What the conventional presentation omits is the horizon at which cumulative forecast error can exhaust the forebay band (3.5). It also requires care when comparing efficiency metrics across locations because their values depend on observed variance (1.6). The approaches agree on the overall product ranking; the tailored metric expresses that ranking in units tied to a GRH constraint.

3.5 Advantage of tailored metrics

(§3.1 Tailored metrics) Please explain the advantage of introducing tailored metrics. For example, it would be helpful to show a case where using tailored metrics led to a different forecast onclusion compared to general metrics.

The generic and tailored metrics do not select different products: forecast B outperforms A on both. The tailored metric adds the decision horizon. Under the idealized neutral-start stress test, exceedance probabilities over the first 24 hours of each product’s horizon are lower for B than for A at all three dams: 31.0% versus 56.6% at Wilder, 47.7% versus 67.9% at Bellows Falls, and 53.8% versus 71.1% at Vernon. By 32 hours, the longest horizon the three products share, B reaches 42.8–65.5% while A reaches 61.2–82.0%. Product B therefore holds a consistent advantage, but passive use of any of the products is untenable: by the end of the first day each leaves the band in roughly a third to three-quarters of issues. NSE, KGE, PBIAS, and correlation do not express this result in forebay units.

On comparison to past studies, the introduction already discussed the head-to-head evaluation exercises (the HEPEX community experiments and the Bureau of Reclamation forecast rodeos), and we have sharpened what distinguishes this work: those evaluate forecast quality in general, while this framework evaluates fitness for one utility’s operating constraints under live conditions.

Reviewer 4

The overall idea is relevant and practically useful for hydropower operators seeking to assess competing forecast products. The paper is generally well-written, and the iterative, operator-inclusive evaluation approach adds value.

Thank you for the careful reading and for the specific, actionable comments. We have addressed all nine. Comment 4.2 led us to check the evaluation period against the long-term record, which corrected a claim we would otherwise have overstated.

4.1 Smoothing window size and sensitivity

Lines 80–81 recommend a triangular smoother for calculated inflows but do not specify the smoothing window size [...] The window size must be stated explicitly, and a sensitivity analysis or justification for the chosen value would strengthen the paper.

The revised methods state the filter and report a sensitivity analysis. We apply two successive 24-hour moving averages to hourly calculated inflow. Their convolution produces a triangular kernel spanning 47 observations. The 0, 6, 12, and 24-hour labels below refer to the width of each component moving average. We selected 24 hours to match the day-ahead volumetric decision. Day-ahead NSE over the four component widths is:

Location	Forecast	0 h	6 h	12 h	24 h
Wilder	A	0.70	0.72	0.72	0.72
	B	0.82	0.83	0.83	0.83
	GRH	0.87	0.87	0.88	0.88
Bellows Falls	A	0.50	0.51	0.52	0.55
	B	0.90	0.90	0.90	0.91
	GRH	0.84	0.83	0.84	0.85
Vernon	A	0.79	0.79	0.79	0.81
	B	0.88	0.88	0.88	0.89
	GRH	0.88	0.88	0.88	0.89

Metrics improve modestly with window width, as expected since smoothing removes observation noise that no forecast can reproduce; the largest change is 0.05 NSE (Bellows Falls, forecast A). The improvement is not strictly monotonic, the in-house forecast at Bellows Falls dipping from 0.84

to 0.83 between 0 and 6 hours before recovering, which we attribute to sampling rather than to any property of the smoother. The ranking is unchanged at every width at every location, and the gap between the commercial products is stable: forecast B leads A by 0.39 NSE at Bellows Falls with no smoothing and by 0.37 with 24-hour smoothing. The 24-hour column reproduces Table 1 exactly, so the sensitivity analysis and the main results come from the same pipeline.

The revised text recommends reporting the filter, testing sensitivity to its width, and matching the component window to the decision timescale.

4.2 Hydrologic representativeness of the evaluation period

Line 116 states the evaluation ran from July 2023 to May 2025. The authors should discuss whether this period captured any notable extreme hydrologic events [...] and how climatologically representative it is relative to the long-term record.

Thanks for pointing this out, this check led to two changes.

First, the stated evaluation period was wrong. The record analyzed runs from June 2023 to April 2025, and the three products do not cover it uniformly: forecast B spans the full period (657 day-ahead forecasts per dam) while forecast A and the in-house forecast both end in November 2024 (539 and 528 respectively). We have corrected the period, added the per-product sample sizes to Table 1, and verified that restricting the comparison to the 510 days on which all three products are present leaves the ranking unchanged at all three dams, with individual metrics shifting by no more than 0.03 NSE. We now report this partial coverage as a characteristic feature of real-time evaluation rather than smoothing over it.

Second, on the events. A new subsection reports that the period captured the July 2023 Vermont flood, which produced the largest calculated inflows of the evaluation at Vernon (83,100 cfs, 2,353 m³/s) and Bellows Falls (67,000 cfs), and a December 2023 rain-on-snow event that produced the largest inflow at Wilder (27,600 cfs), along with spring freshets in 2024 and 2025 and dry summers in 2023 and 2024. Checked against the long-term annual peak series at the USGS reference gages, the July 2023 event ranks among the ten largest flows on record on the main stem, though below the 1936, 1938, and 1927 floods and the 2011 Tropical Storm Irene peak. The period was therefore not hydrologically quiet, but neither did it sample the extreme upper tail, and the manuscript now states that the conclusions are conditional on the observed period. Reviewer 2 asked in comment 2.20 for the limitation to be stated explicitly, which we have added to the limitations discussion.

One caveat: the gage annual peak series are instantaneous peak discharge at the gage, while our values are 24-hour smoothed calculated inflow at the dams, so the two are not strictly commensurable. We have therefore phrased the manuscript statement as an approximate ranking rather than a numerical comparison.

4.3 Characterization of the commercial forecasts

Line 117 [...] the paper should provide at least a general characterization of each forecast's methodology (e.g., physics-based, machine-learning-based, hybrid, statistical).

Agreed, and Reviewers 1 and 2 raised the same point. Please see comment 1.4 for the characterization we have added and for what we cannot disclose. In summary: forecast A is a conceptual rainfall-runoff model with hydraulic routing driven by commercial numerical weather prediction, issued once daily out to roughly 8 days; forecast B is a machine learning model trained on gauge and remotely sensed observations, issued twice daily out to 10 days; the in-house forecast combines RFC forecasts with routing and operator judgment and extends about 32 hours. The issue fre-

quencies and horizons are taken from the forecast archive rather than vendor documentation. We have also added that neither vendor released model internals or calibration diagnostics, which is relevant to the framework’s motivation rather than merely a caveat.

4.4 Extension to probabilistic and ensemble forecasts

The paper focuses only on deterministic forecast evaluation [...] the framework could be briefly extended to probabilistic or ensemble forecasts using metrics such as CRPS, reliability diagrams, or rank histograms.

We agree and have added a methods subsection, “Extension to probabilistic and ensemble forecasts.” Reviewer 2 asked for the same in comment 2.5. It makes the point that the framework is agnostic to whether the forecast is deterministic or probabilistic, since only the metrics and visualizations change and not the process, and notes that CRPS is the natural accuracy measure for ensembles, with the useful property that it reduces to mean absolute error for a deterministic forecast so the two can be compared on a common scale, alongside rank histograms and reliability diagrams for assessing ensemble properties.

Two connections are worth drawing out. The forebay exceedance metric we develop is already a probabilistic statement derived from a deterministic forecast, computed across the empirical distribution of that forecast’s past errors; given an ensemble it would be computed directly from the members, which is both simpler and better founded. And operators bidding into a day-ahead market must submit a single quantity, so a probabilistic forecast must be reduced to a point value at the moment of the bid, which is why improvements in deterministic skill remain directly useful. This is a description of how the framework would extend rather than a demonstration, since all three products evaluated here are deterministic.

4.5 Hyphenation of “real time”

Throughout the manuscript: “real time” should be hyphenated when used as a compound adjective. Usage is inconsistent.

Corrected throughout, including the title. Reviewer 2 raised the same point in comment 2.3, where we describe the treatment.

4.6 KGE citation

(Methods, line 92) Remove the placeholder “(refs)” and insert the actual KGE citation (Gupta et al. 2009).

Corrected. The placeholder is replaced with Gupta et al. (2009), and we have added Nash and Sutcliffe (1970) for NSE in the same sentence. We also corrected the misspelling Reviewer 2 caught in comment 2.9: “King-Gupta Efficiency” now reads “Kling-Gupta Efficiency.”

While checking this we found and fixed a broken citation key elsewhere. The key used in the introduction omitted the trailing “a” in the bibliography key, so that citation was not resolving. All citation keys now resolve against the bibliography.

4.7 RMSE or MAE in Table 1

(Table 1) Consider adding RMSE or MAE values alongside the existing metrics to provide a sense of forecast error magnitude in physical units.

We added RMSE in cfs to Table 1. Forecast A’s day-ahead RMSE at Bellows Falls is 5,904 cfs against 2,615 for forecast B, which conveys the practical size of the difference more directly than

the NSE values of 0.55 and 0.91.

Table 1 also reports the per-product sample size described under comment 4.2.

4.8 Figure 5 caption

(Figure 5 caption) The caption does not specify that the error bars represent the 2.5th and 97.5th percentiles. Figure caption needs to be self-contained.

Corrected. The Figure 5 caption now states that lines show the median and shaded bands span the 2.5th to 97.5th percentiles. It also defines the pass-inflow stress test and the ± 0.5 foot forebay band. We reviewed all captions for self-containment.

4.9 What coupling to scheduling models would entail

(Discussion) The authors note that coupling this evaluation with hydropower scheduling models is an important next step. This point could be strengthened by briefly discussing what such a coupling would entail and what data would be needed.

The original text named the next step without describing it. We now explain that the real-time trial first identifies operationally viable products and the constraints and decision rules that a value study must represent. The selected forecasts would then drive a common scheduling model, and the operating strategy may need revision to make use of a more accurate product.

On the data required, such a coupling would route each candidate forecast through a scheduling model to produce the generation schedule it implies, then value that schedule against realized inflows and market prices, so that forecast differences are expressed in revenue rather than skill. This needs the utility’s scheduling model with its operating constraints and unit efficiency curves, historical day-ahead and real-time market prices at the relevant node, the realized inflow record, and the archive of forecasts as issued so that schedules use only information available at bid time. That last requirement is a further reason a real-time evaluation is a useful precursor, since it produces exactly this archive as a byproduct. The forebay stress test also shows why the compliance constraint cannot be omitted: cumulative forecast error frequently exhausts the operating band under passive use. A value study should admit only compliance-feasible schedules and represent operator response as the band is approached. The same coupling is what a full value assessment (2.1) and an estimate of realized forebay exceedance rates (2.19) would both require.

Summary of changes

Framing. The contribution is now stated as the design of a comparable multi-product operating trial rather than the absence of a hindcast alone (editor, 1.1, 2.2, 3.2). The process now has seven steps, beginning with definition of decisions, constraints, horizons, the incumbent forecast, and the common trial period. The revised discussion explains how a real-time trial identifies operationally viable products and defines the constraints for a subsequent common hindcast and value study (3.4). The abstract was revised to state the same contribution and sequence.

New analysis. The cumulative forebay calculation was corrected and Figures 5 and 6 were regenerated, now indexed by hours into each product’s delivered horizon and truncated at 32 hours. The corrected result identifies a consistent advantage for product B across that window, with all three products leaving the forebay band in a substantial fraction of issues by the end of the first day (3.5). The unsupported non-neutral-start analysis was removed. An NSE skill score considered during revision was also removed because the product and reference samples were not identical (1.6). The manuscript instead interprets NSE with RMSE, sample coverage, and benchmark forecasts.

Smoothing sensitivity was tested over 0, 6, 12, and 24-hour windows (4.1, 2.8). RMSE in cfs, per-product sample sizes, and a common-period check added (4.7, 4.2). Downstream bias propagation quantified (1.5). New subsection on the hydrology of the evaluation period (4.2, 2.20).

New descriptive material. New subsections on operating context (1.3), forecast products (1.4, 2.16, 4.3), and the extension to ensemble forecasts (4.4, 2.5). New tables of basin characteristics (1.5) and tailored metric provenance (3.2). Expanded treatment of feedback mechanics (3.2), iteration timescales (3.3), the scheduling model coupling (4.9), and the pass-inflow assumption (2.19).

Corrections. Evaluation period corrected to June 2023–April 2025 (4.2). NSE values mislabeled as KGE in the cross-site paragraph, now rewritten (1.6). Monthly KGE for forecast A in January and February corrected to 0.45 and -0.67 , and the snowmelt season to March–May (2.18). Pearson correlation denominator corrected (2.13). Bias definition corrected and made sign-consistent with PBIAS (2.12). KGE rationale corrected (2.10). “King-Gupta” $\rightarrow$ “Kling-Gupta,” “(refs)” replaced with Gupta et al. (2009) and Nash and Sutcliffe (1970), and a broken citation key fixed (2.9, 4.6). Knoben et al. (2019) moved adjacent to the threshold it supports (2.11). Persistence redescribed (2.7, 1.2). Capacity clarified as installed nameplate (2.14). Site relationships and selection explained (2.15). “real time” hyphenated attributively throughout (2.3, 4.5). SI units added (2.17). Figure captions made self-contained (1.8, 4.8). Tables rebuilt (1.7).

Requests we were not able to fully satisfy

The following requests remain partly addressed.

- **Climatological benchmark in Figures 3 and 4** (1.2). Recommended in the text but not added: a defensible daily climatology of calculated inflow at the dams needs a long record which is reflective of the current operations.
- **Calibration and validation diagnostics for the commercial products** (1.4). Not released to us by either vendor.
- **A full forecast value assessment** (2.1). We cited the literature and gave a path forward, but no value metrics were computed.
- **Realized forebay exceedance rates** (2.19). Needs the scheduling model coupling of 4.9.
- **Demonstrated probabilistic evaluation** (2.5, 4.4). Described but not demonstrated, as all three products evaluated are deterministic.

The revision corrects the stated evaluation period, monthly values, NSE/KGE label, Pearson correlation equation, and cumulative forebay calculation. It also removes an invalid skill-score comparison introduced during revision. The revised claims now follow directly from calculations that use the published data.